Task-Conditioned Skill-Boundary Parameter Reinforcement Learning for Reliable Multi-Stage Manipulation

A controlled native-physics study and falsifiable method pilot

ARES Research Working Draft

28 October 2026 — not a submission-ready method claim

Abstract

Local completion of a manipulation primitive does not guarantee that the resulting physical state permits its successor to succeed. We study executable parameter adaptation at two physically verified boundaries in native StackPyramid-v1, using a common Panda planner and actual grasp, release, support and stability readbacks. A cold-reset training-design study finds nine cases with different downstream outcomes despite successful predecessor verification, and 18 matched-prefix contexts where the second parameter changes the outcome. We train matched finite-horizon PPO and a paired-regression-risk candidate with equal data, initialization and action authority across three seeds. This draft reports all outcomes, strong fixed/geometry and supervised baselines, matched-budget information ablations and physical cost. The empirical tables below determine the scope of the method claim: opportunity and engineering integration alone are not algorithmic novelty. Experiments use privileged simulator state, not vision or a physical robot.

1 One scientific question

We ask whether a policy can select a physically executable predecessor parameter before release, observe the actual stable released state, and adapt the successor parameter to improve whole-task reliability beyond strong nominal control and ordinary RL with identical authority. The industrial motivation is the separation of perception, Skill Contract/Runtime, parameter policy and robot control in ARES-R. Only the public Panda simulation layer is evaluated here; no JAKA, AMR, point-cloud policy or site deployment is tested.

A waypoint is not a completed skill. A valid handoff must include actual gripper release, retreat, stable objects, support contact and robot state. We therefore separate three claims: physically controllable terminal states exist; a shared executable interface is correct; and a learning mechanism improves independent outcomes. The first two do not establish the third.

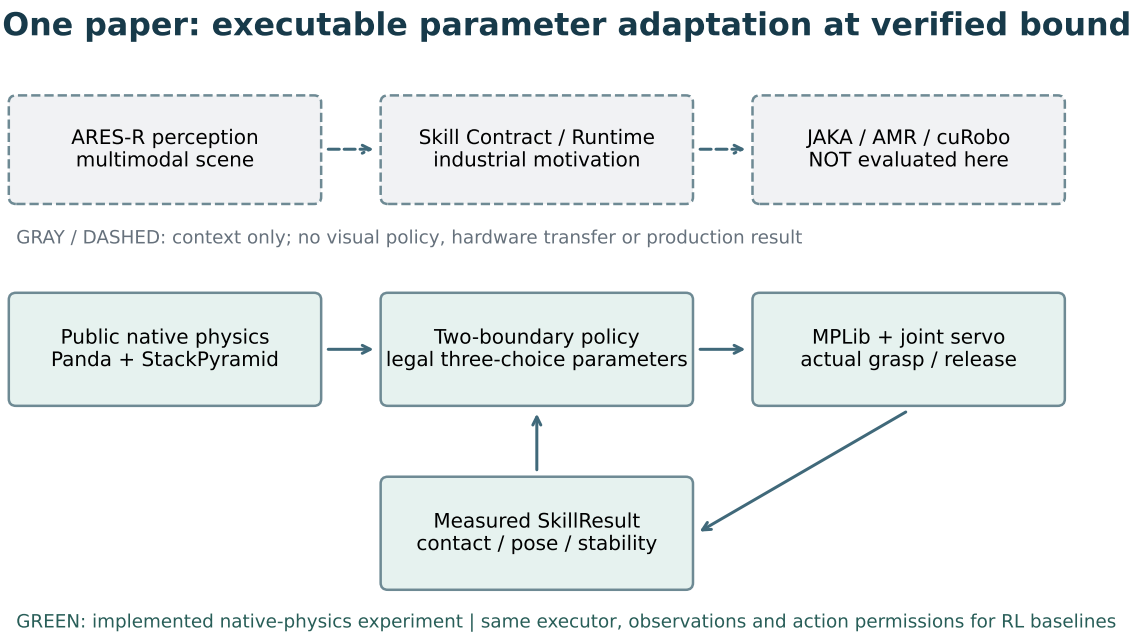


Figure 1: Evaluated simulation components versus industrial context. Dashed components are not experimental achievements.

2 Relation to prior methods

RAPS already learns parameters of manually specified primitives [1]. STAP coordinates geometric dependencies using skill values [2]; surgical ViSkill learns chaining behavior from whole-task value [3]. Foresight Residual RL explicitly optimizes terminal handoff quality with downstream-success predictions [4]. DAWN studies anchored residual value learning and normalization [5]; it is not reproduced by this PPO pilot. We do not claim that downstream value, parameterization, nominal-value subtraction or adding a predictor is new. Any eventual contribution must be a specific independently validated mechanism operating through physically executable contracts.

3 Physical task and common skill contract

We pin public ManiSkill commit 62ff3a5896b4d5b4cf0ac4c8d79afe600c9404a3, StackPyramid-v1 and its original sparse reward, success predicate, physics and 250-control-step horizon. Native control frequency is 20Hz. A red cube A is placed adjacent to green B; blue C must then be placed on the supports. The upstream predicate is used unchanged and evaluated independently at the native horizon.

The common program is: pick A; place, open, retreat and verify A; observe and decide; pick C; place, open, retreat and verify C. No resets, object setters, teleports or cached outcomes occur between skills. The public MPLib planner generates joint paths; the servo consumes every third path sample with a bounded readback correction. Screw planning has two attempts and a 0.5 s RRT fallback. Symmetry-equivalent gripper orientation avoids gratuitous wrist rotations. These are shared engineering controls, not learned gains. Cube obstacles are not encoded in a certified scene-aware motion planner: actual contact and failures occur in native PhysX. This benchmark is not industrial collision certification.

Decision 0 selects an A support-placement direction offset in $\{-\pi/2, 0, +\pi/2\}$ relative to the initial A–B axis. The support clearance accounts for observed B orientation. Decision 1 is requested only after verified release of A and selects a C grasp-yaw offset in $\{-0.35, 0, +0.35\}$rad. The desired C center remains the measured A/B midpoint; actual held-object-to-TCP offset is used for retargeting. Requested controller angles are not treated as measured object rotation.

The boundary requires three consecutive control readbacks with finger opening > 65mm, object not grasped, all three object linear/angular speeds < 0.01 m/s and < 0.5 rad/s, robot joint speed < 0.15 rad/s and XY target error < 15mm. A additionally needs table-height error < 8mm and actual table contact. C needs

target-height error < 8mm and measured contact with A or B. Measured forces and projected footprint overlaps are recorded; projection is not contact area. Native success and verified program completion are reported separately.

All batched episodes physically run all 250 control steps, including terminal holds. Logical completion is a diagnostic, not saved physical time. Candidate joint plans may be reused only when the complete measured per-environment object pose/velocity, robot q/qvel and target match to 2×10^{-6} max absolute, with the same seed, label and chosen-action prefix. Physical servo and effects always run again. Cache reuse and actual planning costs are separated; no algorithm speedup is inferred from cache order.

4 Finite-horizon two-boundary learning

At a visited boundary, o_k contains 57 privileged features: proprioception, current relative object/goal geometry, object orientations, observed position changes, support-contract margins, stage and remaining budget. The actor never receives future outcome labels or case IDs. Full state may make history features redundant; we do not assume a history benefit.

The physical transition is $s_{k+1} = \text{Execute}(\mathcal{S}_k, a_k, s_k)$ followed by independent verification. A failed predecessor terminates the program and creates no virtual second decision token. For finite native horizon $T = 250$, option duration is the difference between actual decision steps, with the final option including terminal hold until T. We explicitly use $\gamma = 1$. Whole-task native terminal success Y is the shared return for each visited decision. There is no reward change favoring the candidate and no claim of physical-time optimization.

Ordinary PPO and the candidate share actor/critic 64-by-64 tanh networks, train-only normalization, initialization, sampled scenes, terminal-return advantages and a common supervised warm start. PPO uses clip 0.2, Adam 3×10^{-4}, eight epochs per 16-episode update, value coefficient 0.5 and entropy coefficient 0.02. Each main model receives 12 updates,192 episodes and 48,000 physical control interactions. Seeds 31,37,43 are all retained.

The conventional candidate tests a frozen paired-regression estimator $h_\phi(o, a)$. Training-only cold-reset labels are $d = \mathbb{1}[Y_{\text{nom}} = 1, Y_a = 0]$: at boundary 0 the suffix is nominal; at boundary 1 nominal is the C choice under the identical A prefix. Binary cross-entropy fits h_ϕ. Its actor objective adds

$$L_{\text{cand}} = L_{\text{PPO}} + \beta \, \mathbb{E}_o \sum_a \pi_\theta(a \mid o) h_\phi(o, a), \quad \beta = 2.$$

Ordinary PPO uses $\beta = 0$ with exactly the same data access and warm start. This heuristic trades predicted downside against return; it is not a calibrated risk guarantee or new policy-gradient theorem. Both actor and critic are trained with actual terminal native outcomes. Predicted risk is never substituted for a physical outcome.

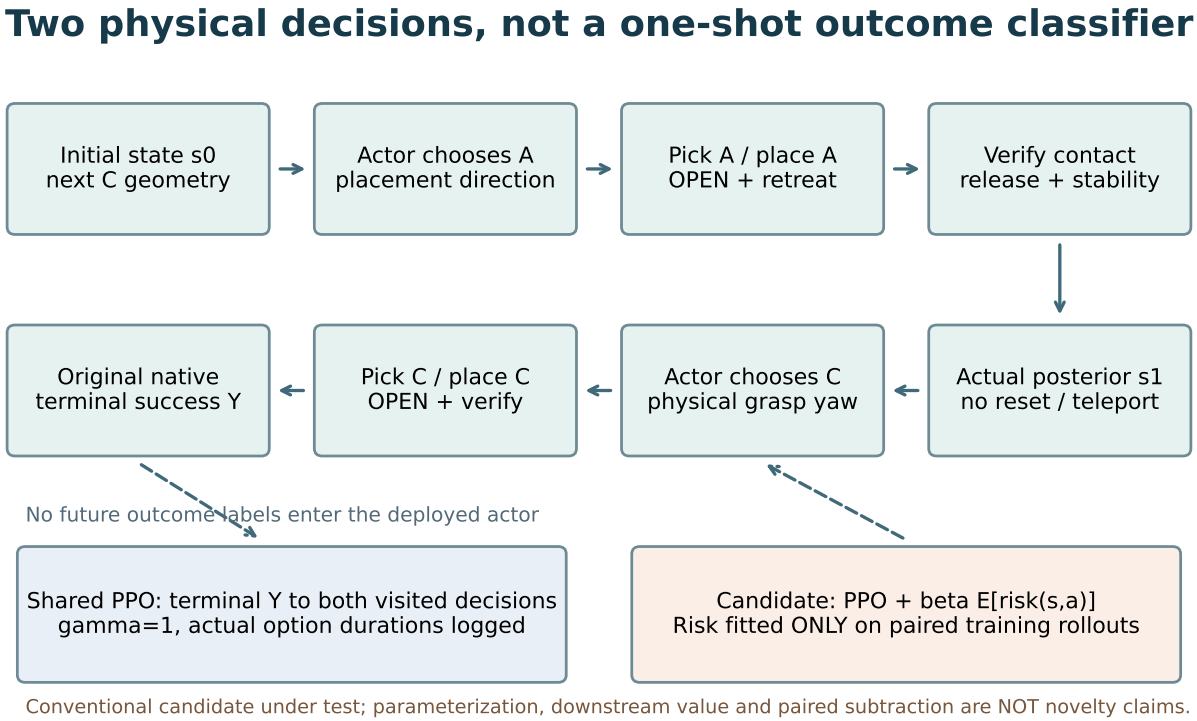


Figure 2: The second parameter is selected after actual release/stability, unlike a one-shot contextual classifier. Failed A has no successor decision.

5 Protocol and fair comparisons

The 64 new training-design scenes 420000–420063 form an exhaustive 3-by 3 cold-reset intervention grid (144,000 control interactions). The first 48 cases, in three reset batches, fit the shared absolute-outcome and paired-risk networks; the last 16 cases, in a separate batch, report calibration without selecting epochs. Both networks use fixed 400-epoch fits. Posthoc audit finds only 9 positive regression labels among 546 eligible risk-fit tokens. On the separate 189-token calibration batch, risk Brier is 0.0233 versus 0.0207 for a constant-prevalence predictor; mean predicted risk is 0.0065 versus observed prevalence 0.0212. This diagnoses a sparse-label calibration failure, not a development-tuned correction. The absolute-result model has training Brier 0.0104 and calibration Brier 0.0779, indicating a nontrivial generalization gap rather than certainty. Online training uses new and reused cases within a separately declared 420000–420511 pool; the 192 actual cases per model are disclosed. Old 16 Pyramid examples and old Stack 64 scores are excluded.

All methods are evaluated on the same 64 new development scenes 520000–520063 with matching initial-state hashes, not only nominal failures. Baselines are strong nominal, a geometry rule choosing A-C and C-jaw clearance, the best fixed pair from the training grid, an absolute-result Q argmax, a paired-risk ranker, matched ordinary PPO and the candidate. Subtracting a state-dependent nominal Q alone cannot alter an argmax; the ranker additionally uses paired downside risk. Finite exhaustive search is the fixed/search baseline; no CEM reproduction is claimed.

Three exploratory ablations remove first-decision next-object geometry, replace second-decision object geometry/effects with the initial snapshot, or commit the second parameter at the initial observation. They train 64 episodes/16,000 interactions, compared against the full candidate at its same-budget update 4. Masked inputs can create distribution shift for the frozen teacher; these are exploratory information tests, not clean proof of a novel posterior-identification mechanism. The zero-risk baseline is ordinary PPO. No best-seed or development-selected checkpoint is reported.

Task scenes and training seeds are distinct statistical units. We report paired rescues/regressions on identical cases and descriptive bootstrap intervals both over cases and over the four 16-environment reset batches, crossed with training seeds. Three seeds and four batches yield limited precision. The sealed 620000–620127 scenes remain unopened pending a reliable increment.

6 Results: physical opportunity versus method benefit

6.1 Interventions change actual downstream outcomes

The nominal training-design pair succeeds 52/64; the best single fixed pair 59/64; the seen candidate union 62/64. The union is an ex-post opportunity ceiling, not a deployable strategy score. Nine scenes have different downstream outcomes across A parameters despite verified predecessor completion. Across 181 valid A-prefix contexts, the three C interventions have exactly equal saved decision 1 input states (maximum absolute discrepancy 0);18 contexts show different outcomes. Thus both parameter boundaries have real physical authority.

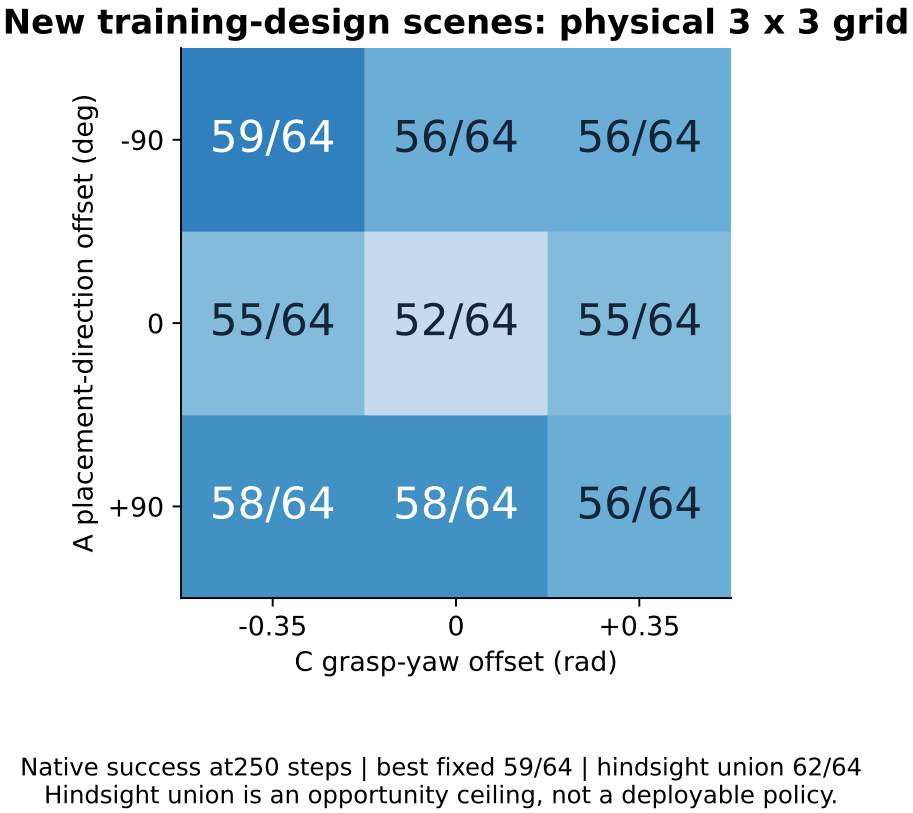


Figure 3: Actual training-design grid, not an RL result. Better training fixed parameters need not generalize.

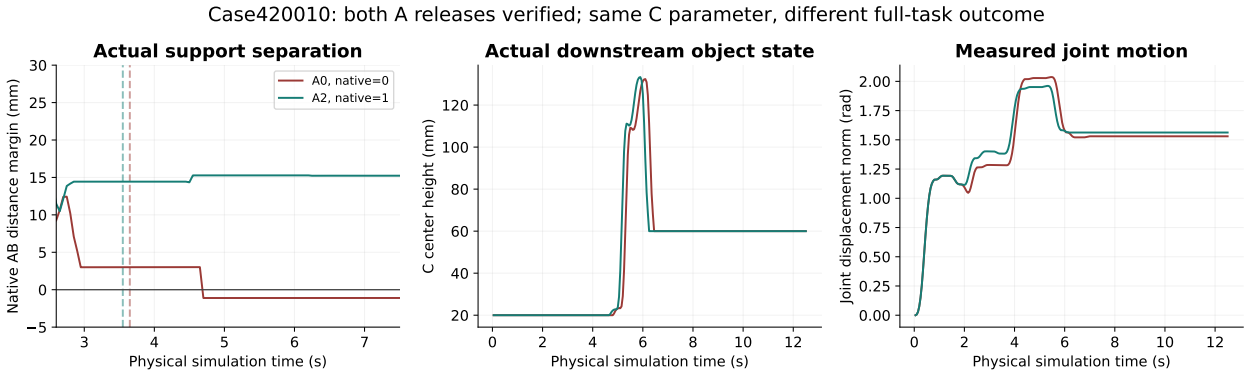


Figure 4: Recorded native physical states in case420010. Both A releases verify; identical C parameter leads to different support separation and final native success. The first panel zooms the support-placement portion. This intervention demonstrates a total physical effect, not isolated causal mediation by margin alone.

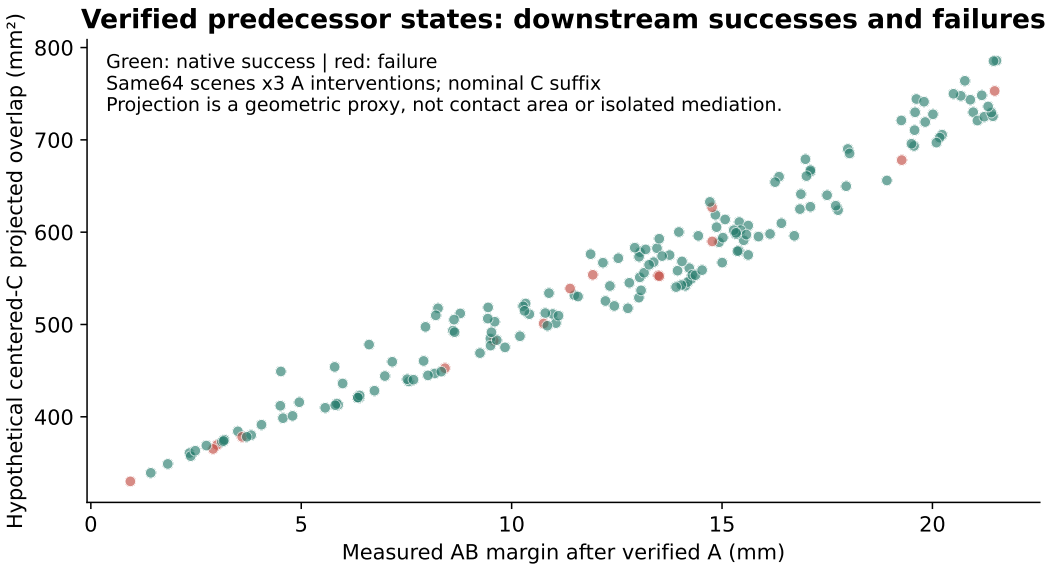
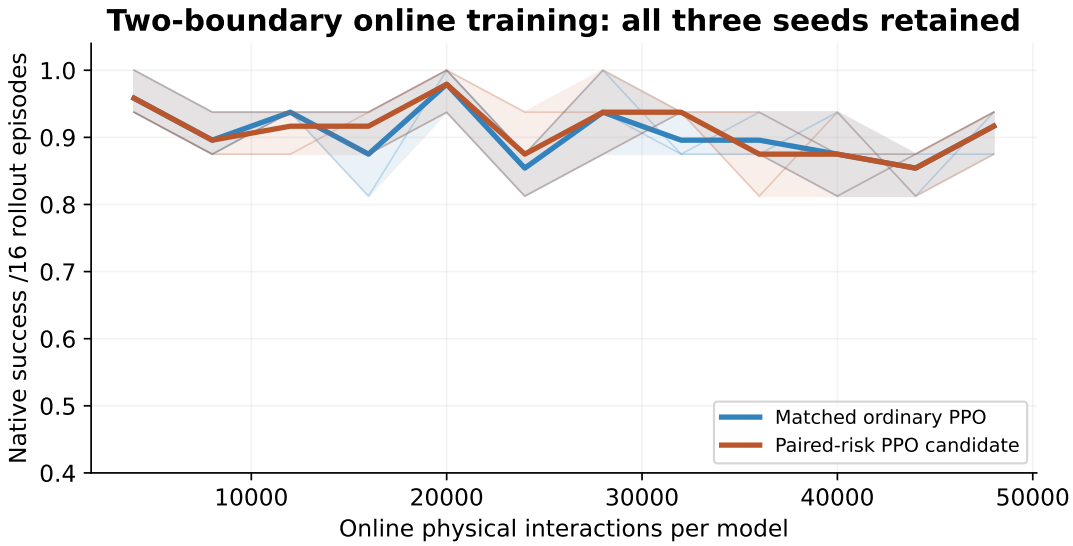


Figure 5: Measured stable handoff geometry and subsequent outcomes. Projected overlap is a proxy; red/green points are not a fitted causal boundary.

6.2 All seeds and same-interface comparisons



Shared144k physical design collection is additional. Training curves are not generalization scores.

Figure 6: Actual online rollouts, all three seeds; lines are not held-out scores. Shared physical design collection is additional cost.

Method	Native/64	Verified/64	A verified	Rescue	Regress
nominal	53	53	57	0	0
rule	55	55	56	4	2
fixed	53	54	55	5	5
absolute_q	53	53	55	4	4
paired_ranker	50	50	54	3	6
ppo_s31_b0_full	51	51	55	4	6
ppo_s31_b2_full	51	51	55	4	6
ppo_s37_b0_full	53	53	56	3	3
ppo_s37_b2_full	54	54	56	4	3
ppo_s43_b0_full	56	56	56	5	2
ppo_s43_b2_full	55	55	57	4	2
ppo_s31_b2_no_next	52	52	54	5	6
ppo_s31_b2_no_posterior	55	56	56	5	3
ppo_s31_b2_no_second	53	53	56	1	1
ppo_s31_b2_full_04	51	51	55	4	6

Table 1: Same64 development cases. Native success and BOTH local boundary checks are distinct; the latter does not imply final native success. Rescue/regress compare each method to nominal on identical cases.

Ordinary same-interface PPO scores 51, 53, 56 out of 64; the candidate scores 51, 54, 55. Candidate minus ordinary averages +0.00 percentage points, with paired seed differences [0, 1, -1] cases. The reset-batch-and-seed bootstrap 95% interval is [-3.12,4.17] percentage points. Native nominal, geometric rule, fixed search, absolute Q and paired ranker score 53, 55, 53, 53 and 50 respectively. These are development results, not sealed-test evidence.

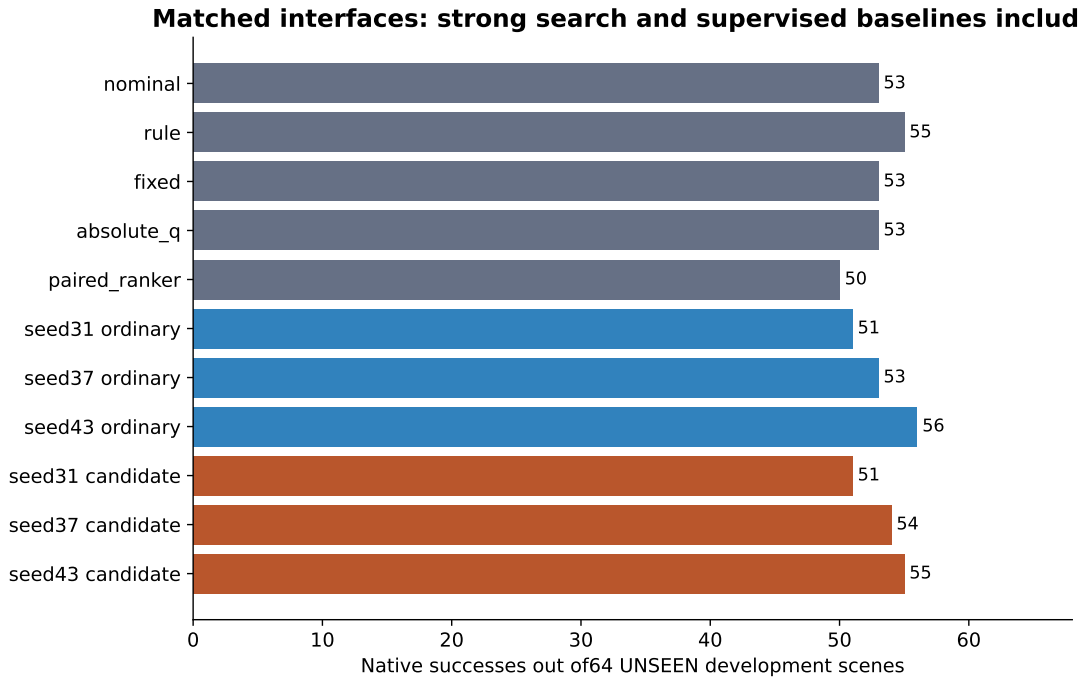


Figure 7: Strong rules, search and supervised baselines are retained; all learning seeds are shown.

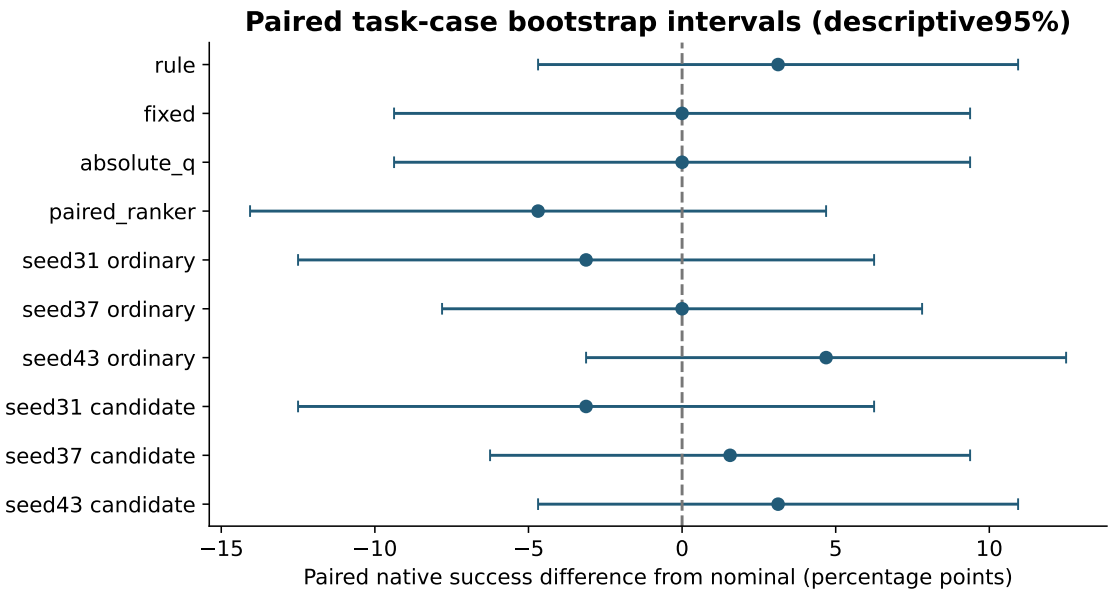


Figure 8: Descriptive paired scene-bootstrap intervals. Reset-batch and training-seed intervals are also recorded in the statistical JSON; neither treats repeated192 evaluations as192 independent task cases.

6.3 Information and physical redecision

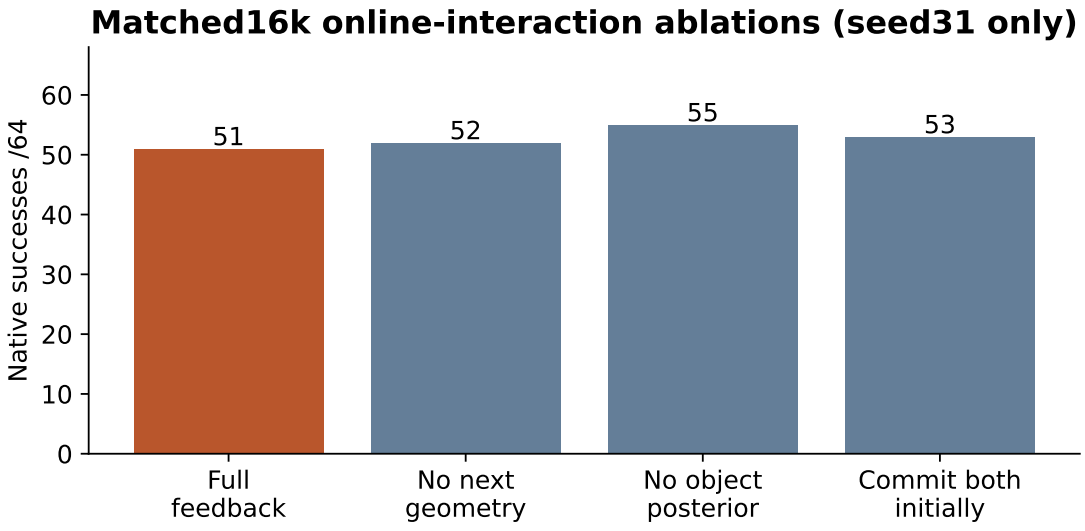


Figure 9: Equal16,000-online-interaction checkpoints. One-seed ablations cannot establish a robust mechanism. Full-state feedback can already contain the physical effect; deleting only effect deltas would be an invalid posterior ablation.

7 Cost, reproducibility and limitations

This round records 786,543 external control interactions:144,000 design collection,288,000 main online training,48,000 ablation training,240,000 development evaluation, plus 66,543 engineering probes/failures. Internal reset settling is excluded from the external-control counter. Every evaluation episode physically executes 250 steps, including terminal padding. No 2x or 3x algorithm speedup is inferred from batched planning or cache order. Per-batch wall time, planner solves/cache hits and logical completion remain separately inspectable

in the cost ledger and raw summaries.

Source, model and scene SHA indexes accompany the tables. Full physical raw trajectories and large videos remain in non-Git evidence storage; selected original measured trajectories and structured training data are versioned. Videos are graphics-only re-renders of saved robot/object states: no idealized motion and no new experiment outcome is inferred. The two qualitative intervention cases were selected from the training-design set, not used as independent method statistics. Isolation uses a network namespace, GPU 0 only and read-only ARES-R mounts.

The study has only one native physical mechanism, privileged state, fixed cube sizes and a small online budget. It does not demonstrate visual perception, zero-shot new-task transfer, sim-to-real or industrial safety. Strong controller corrections are common to every method and are never counted as learning gains. Native fixed-horizon execution costs do not support a wall-time improvement claim. The risk estimator is a small supervised model with limited calibration, and masks may move its inputs off distribution.

8 Research decision and publication scope

The tested risk regularizer does not establish a reliable independent method increment. This is a negative result for this concrete mechanism, not proof that every skill-boundary policy is ineffective. The actual physical opportunity and functioning two-boundary learning interface remain valid assets. We therefore retain the manuscript as a controlled physical study and do not expand to saturated StackCube, weak Peg, five seeds or sealed testing simply to obtain a positive score.

The stable paper object remains executable skill-parameter adaptation across verified physical boundaries. A positive method claim would require reliable gain beyond matched ordinary RL and the best non-RL alternative, a controlled mechanism ablation, additional learning seeds, one-time sealed testing, a distinct physical task and perception robustness. Until those tests pass, this document is a coherent experimental manuscript and falsifiable research record, not a completed SCI method paper. Negative results are retained rather than replaced with an easier baseline or selected successful videos.

References

[1] M. Dalal, D. Pathak, and R. Salakhutdinov. Accelerating Robotic Reinforcement Learning via Parameterized Action Primitives. NeurIPS34,2021. https://proceedings.neurips.cc/paper_files/paper/2021/hash/b6846b0186a035fcc76b1b1d26fd42fa-Abstract.html.

[2] C. Agia, T. Migimatsu, J. Wu, and J. Bohg. STAP: Sequencing Task-Agnostic Policies. ICRA2023, pp.7951–7958. doi:10.1109/ICRA48891.2023.10160220.

[3] Tao Huang, Kai Chen, Wang Wei, Jianan Li, Yonghao Long, and Qi Dou. Value-Informed Skill Chaining for Policy Learning of Long-Horizon Tasks with Surgical Robot. IROS2023. https://arxiv.org/abs/2307.16503.

[4] Y. Liu, X. Zhang, L. Liu, and A. Boularias. Foresight Residual RL for Long-Horizon Robot Manipulation with Vision-Language-Action Models. arXiv:2607.16506,2026 (author-listed IROS2026 acceptance). doi:10.48550/arXiv.2607.16506.

[5] G. Ma, L. Li, H. Wang, Z. Liu, P.-L. Bacon, and D. Tao. What Makes Value Learning Efficient in Residual Reinforcement Learning? ICML2026, PMLR306:84569–84601. https://proceedings.mlr.press/v306/ma26s.html.